Spatially-Aware Forecasting of District-Level Retail Food Prices in Bangladesh Using Neighbouring District Prices

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***Abstract*—Food prices in Bangladesh are not the same everywhere. The price of rice or chicken in one district can be quite different from the price in another, and nearby districts often rise and fall together because they trade with each other. Most machine learning studies on Bangladeshi food prices, however, forecast one national price series and do not use this connection between districts. In this work we test whether the prices of neighbouring districts can help forecast the price in a district. We collected real monthly retail prices of chicken, rice and fish for 63 districts from the World Food Programme and the Department of Agricultural Marketing, cleaned and matched them, and built a district neighbour list for the whole country. We then compared simple baselines with Random Forest and XGBoost models, with and without neighbour-based features, using a rolling test that only uses information available at the time of each forecast. For next-month forecasts, simply repeating the current price was very hard to beat, and the neighbour features did not help. For two-month forecasts, however, the neighbour features reduced the XGBoost error by about ten percent and gave the lowest percentage error of all the models we tested. We also found that a district's price changes follow the average of its neighbours much more closely than those of a distant district. The work is deployed as a web application that gives price forecasts and cross-district price estimates.**

***Index Terms*—food price forecasting, spatial features, neighbouring districts, XGBoost, Bangladesh, WFP**

I. Introduction

Rice, chicken and fish are everyday foods in Bangladesh, and a sudden rise in their price affects millions of families and small traders. Good price information also helps governments and aid agencies decide where help is needed, especially when survey data are not available [1].

One thing that is easy to miss is that there is no single "Bangladesh price". Each district has its own markets, transport costs and supply. At the same time, districts that share a border trade with each other, so a shortage or a price jump in one district often spreads to the next one. Economists have shown this kind of market connection for rice in Bangladesh [2], but machine learning studies on Bangladeshi food prices usually forecast a national or single-market series and ignore it [3]–[7].

For our literature review, our team studied 25 research papers on food and agricultural price prediction in detail. All of them are cited in this paper; the remaining references are the methods, tools and data sources that we used. From this review we found two research gaps. The first is that price forecasting and price anomaly detection are almost always studied separately; only two of the 25 papers did any anomaly detection [8], [9]. The second is that forecasting models for Bangladesh treat each commodity as one national series and do not use the relationship between neighbouring markets. In this paper we work on the second gap. The first gap is left for future work.

Our main question is simple: does adding the prices of neighbouring districts help forecast the price in a district, compared with using only that district's own price history? The main contributions of this work are:

- a cleaned dataset of real monthly district prices from two public sources, with the steps used to match them;
- a neighbour list for all 64 districts and a set of simple spatial features built from it;
- a fair comparison of models with and without these features, including a naive baseline, which six of the 25 reviewed papers did not report;
- a web application where users can see forecasts and estimate prices across districts.

II. Related Work

A. Machine Learning for Crop and Food Prices

Tree-based models are popular for price data stored in tables. XGBoost [10] gave the lowest error for chilli and shallot prices in Indonesia [11], for yearly rice production in Bangladesh [12] and for joint yield and price forecasting in India [13]. A mix of XGBoost and gradient boosting worked best on WFP prices in Kenya [14], while Random Forest beat XGBoost, CatBoost and SARIMA on monthly prices in Nigeria [15]. Deep models such as GRU, BiLSTM and combined LSTM networks did well on long daily series [16]–[20], and grey models were used for monthly pork prices in China [21].

These results do not always hold. ARIMA beat LSTM during the Russia–Ukraine price shock [22], accuracy dropped a lot for 12-month forecasts and during crises [23], and a review of 50 studies found that no single model is best everywhere [24]. Missing values and poor data quality are common problems in developing countries [25], [26].

B. Studies on Bangladesh

AgriPriceBD is a recent daily price dataset for five commodities [27]. Its benchmark study found that simply repeating the last price (the naive model) was the strongest model for several commodities, and that Transformer models performed badly on such small data [3]. On WFP monthly data, a small student network trained by knowledge distillation reached a low error while running on a normal CPU [4], and Random Forest and XGBoost were compared with ridge regression [7]. Other work modelled food inflation for perishable and non-perishable foods [5] and the national food price index [6]. All of these forecast national, single-market or pooled series. None of them uses the link between neighbouring districts.

C. Spatial Information and Anomalies

Only two reviewed papers looked at unusual prices: one combined a Transformer forecaster with an LSTM autoencoder in Indonesia [8], and one found spatial anomalies in electricity prices by splitting the network into zones [9]. Table I compares twelve of the reviewed studies with our work. It shows that most studies report a baseline, but only one Bangladeshi study uses location-based features, and none of them gives the model a neighbour structure between markets.

III. Data

A. Data Sources

We used three public sources:

- **WFP Food Prices for Bangladesh** [28], from the Humanitarian Data Exchange. It has monthly retail prices for many markets and districts. We kept retail prices in taka per kilogram from January 2024 for broiler chicken, coarse rice and live pangasius, because DAM also reports these items.
- **DAM monthly average price reports** [29]. These give a monthly retail price range for eight commodities from June 2025 to September 2026. Only the Dhaka reports are for a single district; the other reports are averages for a whole division.
- **DAM yearly market files**, with one retail price per market for 2025 and 2026 for 65 chicken, rice and fish items in 64 districts. After cleaning they give 1,918 district-year average prices. We use them only for the cross-district estimate in Section IV-E.

B. Cleaning and Matching

All steps were done in one Google Colab notebook so that they can be repeated. First, we removed WFP rows that were not retail prices, not per kilogram, or not for a single district, and changed WFP district names to the DAM spelling (for example, Bogra to Bogura and Chittagong to Chattogram). Second, each WFP item was matched to a DAM item by comparing their price levels. For example, WFP live pangasius was within about 3% of DAM "Pangash (small)" but about 15% higher than "Pangash (big)", so we matched it to the small one. Third, when a district had several markets in the same month, we took the median price, so that one unusual market does not change the district value. For DAM reports we used the middle of the price range.

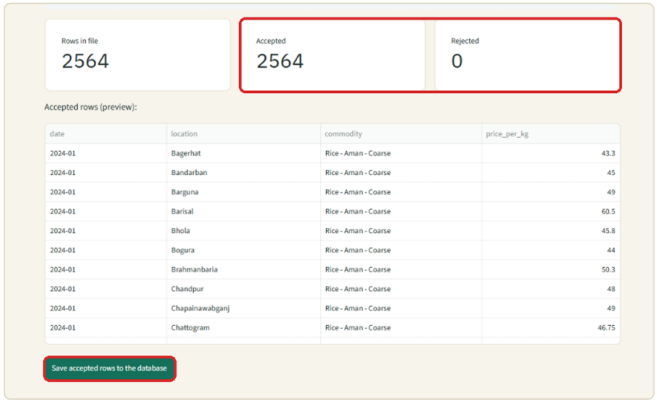


Fig. 1. Uploading the cleaned monthly prices in the admin page of our application. All 2,564 rows passed the checks (Accepted 2,564, Rejected 0).

We removed prices that were more than three times higher or lower than the usual price of that commodity, such as a broiler price of 3,472.74 Tk/kg. Each district series uses only one source, because mixing two sources in one series can create a false jump. Gaps of exactly one month were filled by linear interpolation (167 values, 6.5% of the data); most of these are in November 2025 and January 2026, when WFP did not report most districts, which is before our test period.

We did not copy the division-level DAM prices into each district, because that would make neighbouring districts look exactly the same and would ruin a study about neighbours. We used them as a check instead. When we averaged the WFP district prices for each division, they differed from DAM's division prices by only 2.7% to 4.9% on average, which tells us that the item matching is reasonable.

Finally, the cleaned file was uploaded through the admin page of our application, which checks every row again (known district and commodity names, no future months and a sensible price). Fig. 1 shows this step: all 2,564 rows in the file were accepted and none were rejected, and the preview shows the first rows (coarse rice prices for January 2024).

C. Final Dataset

The final dataset has 2,564 district-month prices in 192 district–commodity series from 63 districts, between January 2024 and September 2026. Table II lists each commodity with its source, the number of districts, the number of monthly prices, the time period and the average price. Coarse rice, broiler chicken and pangasius are the three commodities that are available for most districts, and they make up almost all the data. The other five items come from the DAM reports for Dhaka only. WFP gives 184 of the 192 series and DAM gives the eight Dhaka series.

Fig. 2 shows how many districts have a price in each month. Coarse rice is reported in about 53 to 61 districts during 2024. Broiler chicken and pangasius start in September 2024 with about 50 to 60 districts. From December 2024 the number falls

TABLE I
TWELVE OF THE REVIEWED STUDIES COMPARED WITH THIS WORK (D = DAILY, M = MONTHLY, W = WEEKLY).

Study	Region	Data (freq.)	Models	Baseline	Spatial input
Lestari *et al.* [11]	Indonesia	BPS (M)	LR, RF, XGBoost	Yes	No
Manogna *et al.* [16]	India	AGMARKNET (D)	ARIMA, SVR, XGBoost, LSTM, GRU	Yes	No
Celik and Celik [22]	Global	USDA, ICO (D)	ARIMA, VAR, LSTM, Brownian	Yes	No
Ogol *et al.* [14]	Kenya	WFP (W/M)	DT, RF, XGBoost, LightGBM, stacking	No	No
Odion *et al.* [15]	Nigeria	NBS (M)	RF, XGBoost, CatBoost, SARIMA	Yes	No
Sediyono *et al.* [8]	Indonesia	Bank Indonesia (D)	Transformer, BiLSTM, LSTM-VAE	No	No
Dai *et al.* [9]	USA	PJM nodal (15 min)	t-SNE, DBSCAN, Isolation Forest	No	Zoning
Muhammad *et al.* [3]	Bangladesh	AgriPriceBD (D)	Naive, SARIMA, Prophet, BiLSTM, Informer	Yes	No
Zaman and Alam [4]	Bangladesh	WFP (M)	DLinear, PatchTST, N-BEATS → MLP	Yes	No
Rafid *et al.* [5]	Bangladesh	USAID AVC	RF, XGBoost, stacking	No	No
Shuvo [7]	Bangladesh	WFP	RF, XGBoost, Ridge	Yes	Engineered[a]
Javed [6]	Bangladesh	Food price index (M)	SARIMAX, ANN, Prophet, LSTM	Yes	No
This work	Bangladesh	WFP + DAM (M)	Naive, MA, Ridge, RF, XGBoost (with and without spatial)	Yes	Neighbour list

[a] Location-based features as described by the author; no neighbour structure between districts is reported.

TABLE II
REAL MONTHLY RETAIL PRICE DATA USED IN THIS STUDY (TK/KG). DISTR. IS THE NUMBER OF DISTRICTS AND ROWS IS THE NUMBER OF MONTHLY PRICES.

Commodity	Source	Distr.	Rows	Period	Mean ± SD
Broiler chicken	WFP+DAM	61	719	2024-02–2026-09	174.7 ± 16.9
Sonali Chicken	DAM	1	16	2025-06–2026-09	308.9 ± 26.5
Rice - Aman - Coarse	WFP+DAM	63	1258	2024-01–2026-09	50.9 ± 3.9
Rice - Aman - Medium	DAM	1	10	2025-12–2026-09	57.0 ± 1.1
Rice-Boro-Medium	DAM	1	7	2025-06–2025-12	61.6 ± 1.8
Pangash (small)	WFP+DAM	63	522	2024-02–2026-09	193.2 ± 32.3
Pangash (big)	DAM	1	16	2025-06–2026-09	191.7 ± 7.3
Telapia/Nilotica	DAM	1	16	2025-06–2026-09	209.4 ± 8.4

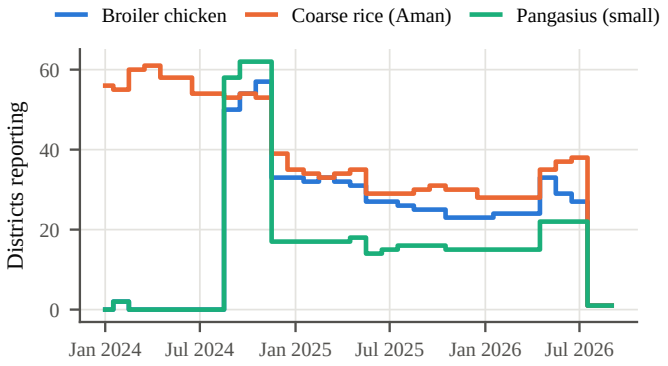
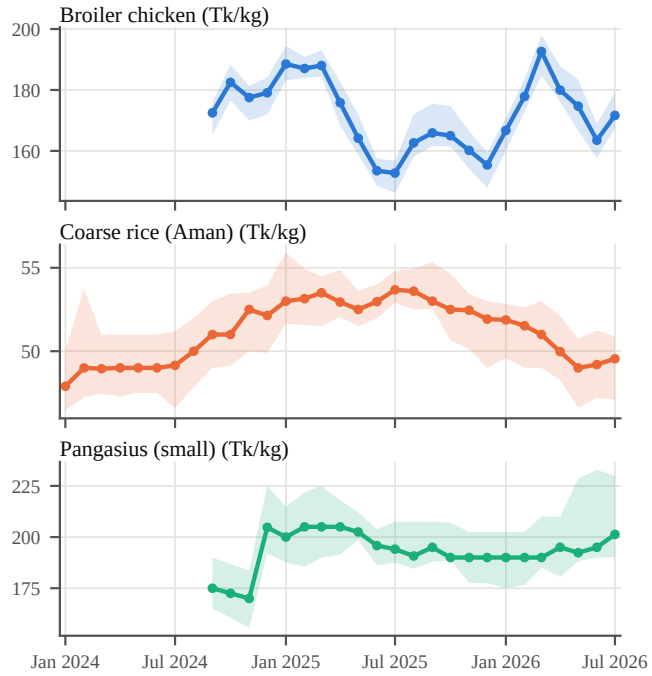


Fig. 2. Number of districts with a price in each month for the three main commodities.



Line: median across reporting districts; band: inter-quartile range (months with ≥5 districts).

Fig. 3. Monthly retail prices of the three main commodities. The line is the median across reporting districts and the band covers the middle half of the districts (25th to 75th percentile). Only months with at least five districts are shown.

to roughly 14 to 39 districts per commodity and rises again to between 22 and 38 districts in mid 2026. After July 2026 only the Dhaka reports from DAM are left, which is why our test period ends in July 2026.

Fig. 3 shows the median price across districts in each month (line) and how much the districts differ from each other (band). Broiler chicken changes the most: its median moved between about 153 Tk/kg (July 2025) and 193 Tk/kg (March 2026). Coarse rice rose slowly from about 48 to 54 Tk/kg between January 2024 and mid 2025 and then fell back to about 49 Tk/kg in mid 2026. Pangasius stayed between about 190 and 205 Tk/kg from December 2024 onwards. Its jump in December 2024 is partly because the set of reporting districts changed at that time (from 62 to 17 districts), so it should not be read as a national price jump. The wide bands show that prices differ a lot between districts, which is the main reason for this study.

IV. METHODOLOGY

A. System Overview

Fig. 4 shows the full system from top to bottom. The three data sources are cleaned in Colab, uploaded through the admin

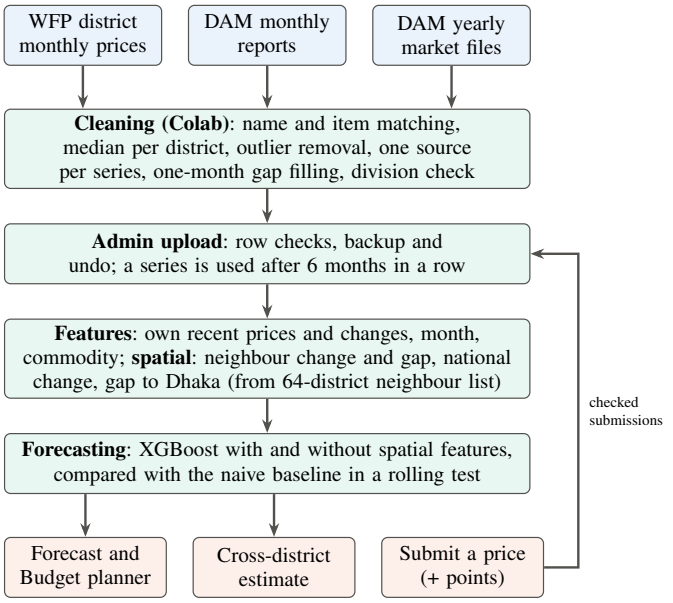


Fig. 4. Overview of the system, from data collection (top) to the web application (bottom).

page and stored. The features are then built for each district and month, the forecasting models are trained and tested, and the results are shown in the web application. Prices that users submit go back to the stored data after they are checked.

The admin page checks every uploaded row, keeps a backup and allows undo. A district series is used for forecasting once it has at least six months in a row, because the model needs the last few months without gaps. For district–commodity pairs that do not have this yet, the application uses a reference series made from the DAM yearly district averages and labels it in the interface; these pairs are not used in any experiment in this paper. A new model can be trained from the admin page. It is first tested on the latest months against the naive baseline and only replaces the current model after the admin confirms.

The web application is built with Streamlit and has five tabs: *Forecast* (prices for 1 to 18 months ahead), *Cross-district* (estimate the price in one district from a known price in another), *Budget planner* (which items may rise most within a budget), *Submit a price* (users add the current price they see and earn points) and *Admin*. A submitted price that is very different from the last known price is held for the admin to check.

B. Forecasting Task

In simple words, the model stands at the end of one month, looks at what is known about a district at that time, and guesses the price of the same item in the same district one, two or three months later. We call the number of months ahead the *horizon* h.

For example, at the end of March 2026 we know the broiler chicken prices of Dhaka up to March. A forecast with $h = 1$ predicts the April price, $h = 2$ predicts May and $h = 3$ predicts June. In all three cases the model may only use prices up to March, never prices from April onwards.

Each row in our training data is one district, one commodity and one month. It holds the features for that month (Section IV-C) and the true price h months later, which the model learns to predict. We write the price of commodity c in district d in month t as $p_{d,c,t}$, so the target of a row is $p_{d,c,t+h}$.

The previous months are matched by calendar date. If a district has no price for February, its January price is not used as the "previous month" of March; that row is simply left out. We only use rows where the current month and the three months before it all have a price, so every row has a complete recent history.

C. Features

Table III lists all the features and which models use them. They are in three groups.

Own-history features: these describe the district's own recent prices. The level models use the price of the current month and of the three months before it. The change models use monthly changes instead of prices. A change is written as a log-return, $r_t = \ln(p_t/p_{t-1})$, which is almost the same as the percentage change. For example, a price going from 50 to 52 Tk/kg gives $r_t = \ln(52/50) = 0.039$, or about 4%. The change models use the last three changes and the distance of the current price from its three-month average, which shows whether the price is above or below its recent level.

Calendar and item features: the calendar month (1 to 12) lets the model learn seasonal patterns, for example around the rice harvest. The commodity is given as one-hot columns (one 0/1 column per commodity), so one model can be trained on all items together. The level models also get the district as one-hot columns, as in our application.

Spatial features: these describe the other districts. We made a neighbour list for all 64 districts of Bangladesh from the district map: two districts are neighbours if they share a border. The list has 282 pairs, written in both directions. It is fixed and is not chosen from price data, so it cannot leak information from the prices we try to predict. From it we built four features for each district and month: the average change of the neighbours, the gap between the district's price and its neighbours' average price, the average change of all other districts (national change), and the gap between the district's price and the Dhaka price, since Dhaka is the main wholesale market. For example, if a district's rice costs 52 Tk/kg and its neighbours cost 50 Tk/kg on average, the gap to neighbours is $\ln(52/50) = 0.039$, so it is about 4% more expensive than its neighbours.

Only neighbours that sell the item are counted. If no neighbour sells it, the value is left empty and XGBoost handles it on its own, so the models with and without spatial features are trained on exactly the same rows. Spatial values were available for 81% of the test forecasts.

D. Models

We compared seven models:

TABLE III
FEATURES USED BY THE MODELS. L = LEVEL MODELS (RF-L, XGB-L); C = CHANGE MODELS (RIDGE-R, XGB-R, XGB-RS); S = ONLY THE SPATIAL MODEL XGB-RS.

Feature	Meaning	Used by
Own history		
Current price	p_t, price this month (Tk/kg)	L
Previous prices	p_{t-1}, p_{t-2}, p_{t-3}	L
Last change	$r_t = \ln(p_t/p_{t-1})$	C
Earlier changes	r_{t-1}, r_{t-2}	C
Distance from 3-month average	$\ln\left(p_t/\text{mean}(p_t, p_{t-1}, p_{t-2})\right)$	C
Calendar and item		
Calendar month	1 to 12	L, C
Commodity	one-hot, one column per item	L, C
District	one-hot, one column per district	L
Spatial		
Neighbour change	average r_t of the bordering districts	S
Gap to neighbours	$\ln(p_t/\text{neighbours' average price})$	S
National change	average r_t of all other districts	S
Gap to Dhaka	$\ln(p_t/\text{Dhaka price})$	S

- **Naive**: the next price is the same as the current price.
- **MA3**: the average of the last three months.
- **Ridge-R**: ridge regression that predicts the change $\ln(p_{t+h}/p_t)$.
- **RF-L**: Random Forest (300 trees) that predicts the price directly.
- **XGB-L**: XGBoost that predicts the price directly. This is the model used in our application.
- **XGB-R**: XGBoost that predicts the change $\ln(p_{t+h}/p_t)$.
- **XGB-RS**: the same as XGB-R, plus the four spatial features.

The change models turn the predicted change back into a price by multiplying it with the current price. XGB-R and XGB-RS are our main pair. They have the same settings, the same rows and the same test, so any difference between them comes only from the spatial features. All XGBoost models use 200 trees, a maximum depth of 4 and a learning rate of 0.05. We kept the trees shallow because the dataset has only a few thousand rows. Each horizon h gets its own model, because forecasting step by step would also need the future prices of every neighbour.

E. Cross-District Estimate

The Cross-district tab estimates the price in a target district u from a known price in a source district s: $\hat{p}_u = p_s \times \overline{(p_u/p_s)}$, where the price ratio is averaged over the years in which both districts have a DAM yearly price. For example, if Dhaka has usually been 5% more expensive than Chattogram, a Chattogram price of 100 Tk/kg gives an estimate of 105 Tk/kg for Dhaka.

F. Evaluation Method

We used a rolling test. Each month from February to July 2026 was used as a test month. For each one, every model was trained again using only rows whose target month was not later than the month in which the forecast is made, so the model never sees a price from the future. This gave about 400 test forecasts for each horizon from 70 district series.

TABLE IV
TEST RESULTS ON REAL PRICES. FOR $h = 1$: MAE AND RMSE IN TK/KG, MAPE IN %, RMAE COMPARED WITH NAIVE, AND DM TEST p-VALUE AGAINST NAIVE. THE LAST TWO COLUMNS GIVE THE MAPE (%) FOR $h = 2$ AND $h = 3$. THE BEST VALUE IN EACH COLUMN IS IN BOLD.

	$h = 1$ ($n = 397$)					$h = 2$	$h = 3$
Model	MAE	RMSE	MAPE	rMAE	DM p	MAPE	MAPE
Naive	**5.95**	**11.26**	**3.60**	1.000	–	5.42	6.52
MA3	7.78	13.65	4.76	1.307	<0.001	5.96	**6.48**
Ridge-R	6.57	11.76	3.94	1.105	<0.001	5.95	7.70
RF-L	6.80	12.46	4.15	1.143	0.002	5.69	6.79
XGB-L	6.45	11.75	3.95	1.084	0.087	5.50	6.66
XGB-R	6.32	11.51	3.75	1.062	0.171	5.92	7.37
XGB-RS	6.60	11.73	3.96	1.109	0.012	**5.31**	7.12

We report the mean absolute error (MAE) and root mean squared error (RMSE) in Tk/kg, the mean absolute percentage error (MAPE), and the MAE relative to the naive model (rMAE, where a value below 1 means better than naive). To check whether a model is really different from the naive model, we used the Diebold–Mariano (DM) test [30]; a p-value below 0.05 means the difference is unlikely to be due to chance. We also used SHAP values [31] to see which features the spatial model relies on.

V. RESULTS AND EVALUATION

A. Overall Accuracy

Table IV gives the results of all seven models. For next-month forecasts ($h = 1$) the naive model had the lowest error, with an MAE of 5.95 Tk/kg and a MAPE of 3.60%. XGB-R came second (3.75%), but the DM test shows that this small difference is not significant ($p = 0.17$). The moving average, ridge regression and Random Forest were clearly worse than the naive model ($p < 0.01$). Monthly retail prices change very little from one month to the next, so repeating the current price is already a good forecast. The AgriPriceBD benchmark found the same for Bangladeshi daily prices [3].

Fig. 5 shows the same next-month MAPE values as bars, sorted from best to worst. Every other bar ends to the right of the dashed naive line, which makes it easy to see that no model beat the naive model at this horizon. The three XGBoost models are all within 0.4 percentage points of naive, while MA3 is clearly the weakest.

For longer forecasts the picture changes. Fig. 6 shows the MAPE of the naive model and the three XGBoost models for one, two and three months ahead. All models make bigger errors further ahead, as expected. At two months, XGB-RS had the lowest MAPE of all models (5.31% against 5.42% for naive), and its line falls below the naive line. At three months, MA3 (6.48%) and XGB-L (6.66%) were close to the naive model (6.52%), while the change-based models were weaker.

B. Results by Commodity

The commodities behave differently, as shown in Table V and Fig. 7. Broiler chicken changes the most from month

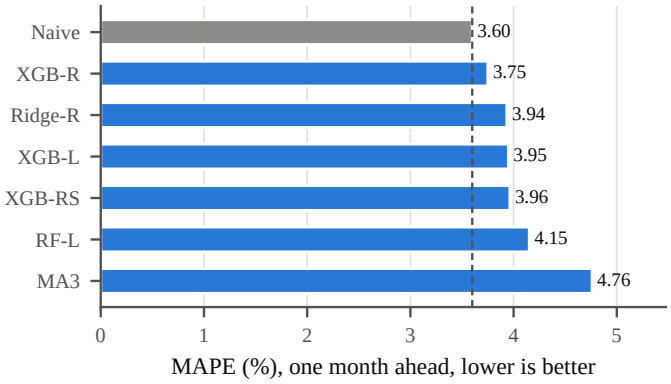


Fig. 5. Next-month MAPE of each model, sorted from best to worst. The grey bar and the dashed line show the naive model.

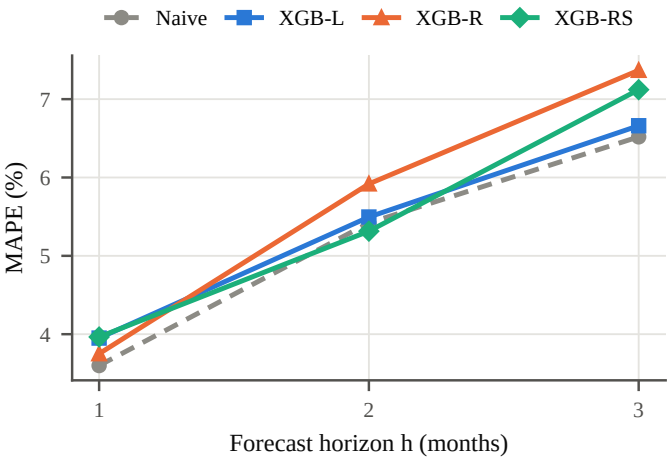


Fig. 6. MAPE for forecasts one, two and three months ahead for the naive model (dashed) and the three XGBoost models.

to month, and here XGBoost did help: XGB-L lowered the MAPE from 6.73% to 5.91%. Rice and pangasius prices often stay the same for several months (Fig. 3), so for them the naive model is very hard to beat. On these two items the change-based models (XGB-R and XGB-RS) stayed closer to the naive model than the level model did, for example 1.82% for XGB-R against 2.28% for XGB-L on rice.

C. Price Movement of Neighbouring Districts

Before looking at the spatial forecasts, we checked whether neighbouring districts really share price movements. For each district and month we compared its price change with (i) the average change of its neighbours, (ii) the change of one random neighbour, and (iii) the change of one random district that is not a neighbour. Fig. 8 shows the correlation for each case; a value closer to 1 means the two move together more closely. For broiler chicken and coarse rice, the neighbour average is a much better match ($r = 0.62$ and 0.49) than a district that is not a neighbour ($r = 0.43$ and 0.14). Even a single neighbour is a better match than a single non-neighbour. Pangasius is the exception: its prices move together

TABLE V
NEXT-MONTH MAPE (%) FOR EACH COMMODITY GROUP. n IS THE NUMBER OF TEST FORECASTS. THE BEST VALUE IN EACH COLUMN IS IN BOLD.

Model	Chicken (n=132)	Fish (n=100)	Rice (n=165)
Naive	6.73	**2.56**	**1.72**
MA3	8.16	3.81	2.62
XGB-L	**5.91**	4.12	2.28
XGB-R	6.58	3.20	1.82
XGB-RS	7.01	3.34	1.91

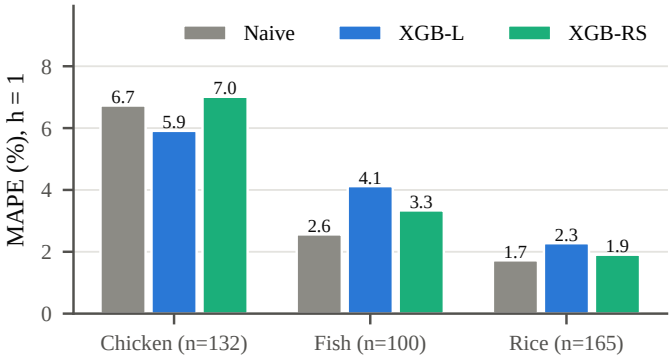


Fig. 7. Next-month MAPE by commodity group for the naive model, XGB-L and XGB-RS. The number above each bar is the MAPE in %.

only weakly (r about 0.2), and being a neighbour makes no difference.

D. Effect of the Spatial Features

Table VI compares our main pair: XGB-R, which uses only the district's own history, and XGB-RS, which also uses the spatial features. For next-month forecasts the spatial features did not help; the MAE was 4.4% higher, and XGB-RS was also significantly worse than the naive model at this horizon ($p = 0.012$). For two-month forecasts the spatial features reduced the MAE by 10.2%, XGB-RS was more accurate than XGB-R in 56% of the cases, and it had the lowest MAPE of all seven models. For three months the gain was small (1.8%).

One possible reason is that a price change in a neighbouring district takes some time to reach the target district. A change that appears in the neighbours this month is more useful for predicting the price two months later than for next month, when the district's own current price already says most of what is needed.

Fig. 9 shows which features the spatial model uses, measured with SHAP. A longer bar means the feature has a bigger effect on the model's predictions. Together the four spatial features (orange) make up about 38% of the model's total feature importance, and the national change is the most important single feature (20.6%). So the model does pay attention to what is happening in other districts, even though this does not lower the next-month error.

E. Cross-District Estimate

We tested the cross-district estimate on the DAM yearly data. We learned the price ratio between two districts from

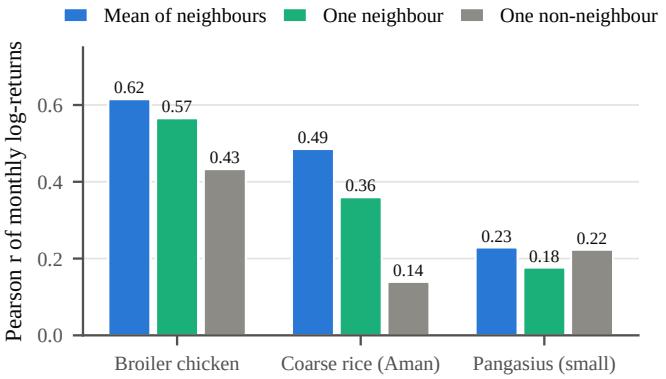


Fig. 8. Correlation between a district's monthly price change and the change of the average of its neighbours, of one random neighbour, and of one random district that is not a neighbour.

TABLE VI
OWN-HISTORY MODEL (XGB-R) AGAINST THE SPATIAL MODEL (XGB-RS). GAIN IS THE REDUCTION IN MAE (POSITIVE MEANS THE SPATIAL MODEL IS BETTER). WINS IS THE SHARE OF FORECASTS WHERE XGB-RS WAS CLOSER TO THE TRUE PRICE. THE NAIVE MAPE IS GIVEN FOR REFERENCE.

		MAE (Tk/kg)		MAPE (%)		Gain	Wins	Naive
h	n	XGB-R	XGB-RS	XGB-R	XGB-RS	(%)	(%)	MAPE
1	397	6.32	6.60	3.75	3.96	-4.4	45.8	3.60
2	396	9.76	8.76	5.92	5.31	+10.2	56.3	5.42
3	395	11.82	11.61	7.37	7.12	+1.8	50.6	6.52

2025 and used it to estimate each district's 2026 price from the 2026 price of another district. We compared this with simply assuming that both districts have the same price. Table VII shows that using the ratio lowered the error from 10.77% to 7.48% over 24,038 district pairs, and the median error from 7.69% to 4.85%. Neighbouring districts are closer in price to begin with (10.17% against 10.82% without the ratio), but after using the ratio both groups have about the same error. This suggests that the price gap between two districts stays fairly stable from one year to the next.

VI. DISCUSSION

Our results give a mixed but useful answer to the research question. Neighbouring districts clearly move together, especially for chicken and rice, and the spatial model makes heavy use of this information. However, this does not automatically lead to better next-month forecasts, because monthly retail prices change slowly and the current price is already a strong guide. The spatial features become useful when we look two months ahead.

This gives two practical lessons. First, every price forecasting study should report a naive baseline. Six of the 25 papers we reviewed did not, and without it a model can look good even when it is not better than repeating the last price. Second, the best model depends on the commodity and on how far ahead we forecast. A real system could choose the model for each case using the same rolling test that our admin page runs before a new model is installed.

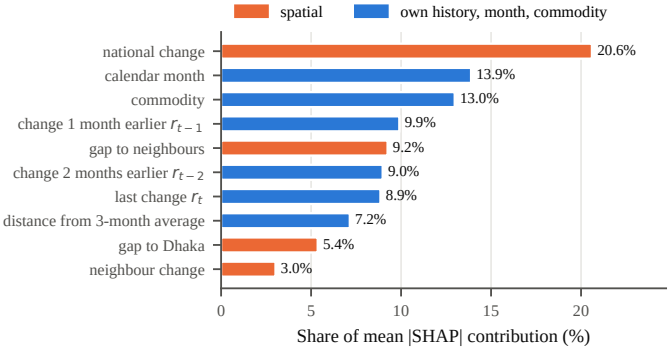


Fig. 9. Share of feature importance (mean absolute SHAP value) in XGB-RS for next-month forecasts. Spatial features are shown in orange and the district's own features in blue.

TABLE VII
ERROR (%) OF THE CROSS-DISTRICT ESTIMATE ON DAM YEARLY PRICES FOR 2026. "SAME PRICE" ASSUMES BOTH DISTRICTS HAVE THE SAME PRICE; "RATIO" USES THE 2025 PRICE RATIO.

		MAPE		Median error	
Pairs	n	Same price	Ratio	Same price	Ratio
All pairs	24,038	10.77	7.48	7.69	4.85
Neighbouring pairs	1,796	10.17	7.53	7.06	4.86
Non-neighbouring pairs	22,242	10.82	7.48	7.75	4.85

All models train in a few seconds on a normal laptop, and the spatial features are easy to explain: "what did the neighbours do last month?" This makes the approach suitable for a small team or an agency with limited computing resources.

The study has some limitations. Our real data cover at most 33 months, the number of districts reporting changes over time, and only three commodities are available for most districts. About 6.5% of the values were filled in for single missing months. The DM test treats all forecasts as independent even though they come from related series. Finally, the neighbour list is based on borders only and does not consider roads or trade routes.

VII. CONCLUSION AND FUTURE WORK

In this paper we studied whether the prices of neighbouring districts can help forecast retail food prices in Bangladesh, which addresses the second gap found in our literature review. Using real monthly prices from WFP and DAM for 63 districts, we found that neighbouring districts share their price movements much more than distant ones. For next-month forecasts, the naive model remained hard to beat and the spatial features did not help. For two-month forecasts, the spatial features lowered the XGBoost error by about 10% and gave the lowest percentage error of all models. We also built a web application that provides these forecasts and cross-district price estimates.

In future we plan to work on the first gap from our review by adding price anomaly detection on top of the forecasts, for example by flagging prices that are far from both the forecast and the neighbouring districts. We also want to collect

longer data for more commodities, use trade routes instead of borders to define neighbours, and add weather and exchange rate information.

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